Chapter 1

The Normal Law of Error

> The theory of probabilities is at bottom nothing but common sense reduced to calculus.
> — Laplace, *Essai philosophique*

When an astronomer measures the same angle a hundred times, the readings scatter about a centre, the small errors many and the large ones few. Gauss asked which law of error makes the arithmetic mean the most probable value of the quantity measured, and found it.[1] If the readings are $x_1, \cdots, x_n$ and their errors are independent, the law must be

$$\varphi(x) = \frac{1}{\sigma\sqrt{2\pi}}\, e^{-\frac{(x-\mu)^2}{2\sigma^2}} \tag{1.1}$$

where μ is the true value and σ measures the spread of the readings.[2] That it is a law at all, that its area is one, rests on an integral known since de Moivre:

$$\int_{-\infty}^{\infty} e^{-x^2}\,dx = \sqrt{\pi}. \tag{1.2}$$

1 Carl Friedrich Gauss, *Theoria motus corporum coelestium in sectionibus conicis solem ambientium* (Hamburg: Perthes and Besser, 1809), §177.

2 Gauss's own measure was the precision $h = 1/(\sigma\sqrt{2})$; the standard deviation σ is Pearson's, of 1894.

1.1 The integral

Square it, and the product of two integrals is one over the plane, which polar coordinates take apart:

$$\left(\int_{-\infty}^{\infty} e^{-x^2}\,dx\right)^2 = \int_0^{2\pi}\int_0^{\infty} e^{-r^2} r\,dr\,d\theta = \pi. \tag{1.3}$$

The same argument gives every moment of the law. With k a natural number, the even moments are

$$E[(X-\mu)^{2k}] = \sigma^{2k}(2k-1)!!, \tag{1.4}$$

and the odd ones vanish, the law being symmetric about μ. Table 1.1 gives the first few, and figure 1.1 the shape of the curve.

moment	$k=1$	$k=2$	$k=3$
$E[(X-\mu)^{2k}]$	σ^2	$3\sigma^4$	$15\sigma^6$
fraction within $k\sigma$	0.6827	0.9545	0.9973

Table 1.1. The even moments of the normal law, and how much of it lies within k standard deviations of the mean.

1.2 Sums of errors

Why should errors follow this law? Laplace's answer was that an error is the sum of many small ones. If $X_1, X_2, \cdots$ are independent, each with mean μ and variance σ^2, then

$$\lim_{n\to\infty} P\left(\frac{\sum_{i=1}^{n} X_i - n\mu}{\sigma\sqrt{n}} \le z\right) = \int_{-\infty}^{z} \varphi(t)\,dt, \tag{1.5}$$

whatever the law of each, so long as its variance is finite. This is the central limit theorem, and it is why the curve of figure 1.1 is drawn wherever measurements are made.

Setting such mathematics well is an old problem. Kernighan and Cherry made a language for it that reads as one speaks it,[3] and Knuth's

3 Brian W. Kernighan and Lorinda L. Cherry, "A System for Typesetting Mathematics," *Communications of the ACM* 18, no. 3 (1975): 151–57.

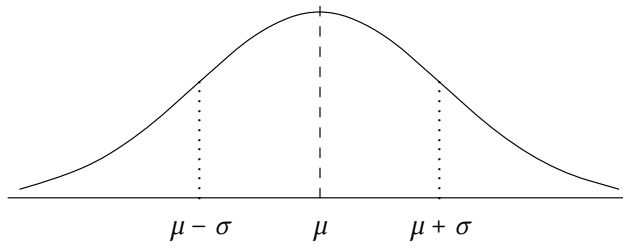


Figure 1.1. The normal law: two thirds of it within one standard deviation of the mean.

rules place each part of a formula as a compositor would.[4] Typography, in Bringhurst's phrase, exists to honour its content.[5]

4 Donald E. Knuth, *The TeXbook* (Reading, Mass.: Addison-Wesley, 1984), ch. 17.

5 Robert Bringhurst, *The Elements of Typographic Style*, 3rd ed. (Vancouver: Hartley & Marks, 2004), 17.